

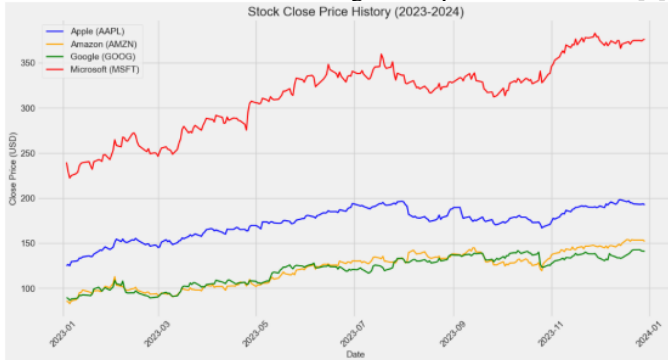
# Comparing RNN and LSTM Models for Stock Price Prediction

Yiheng Yao  
Department of Statistics  
University of Michigan  
Ann Arbor  
Email: yyiheng@umich.edu

**Abstract**—Stock price prediction is a challenging task that has captured the interest of both researchers and investors due to its potential financial rewards. Traditional statistical models, such as ARIMA, have limited capability in capturing complex temporal patterns in financial data. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) models, have demonstrated effectiveness in predicting time series data due to their ability to learn and remember long-term dependencies. This paper aims to compare the performance of RNN and LSTM models in predicting the closing prices of major technology companies, focusing on Amazon, Apple, Microsoft, and Google. Numerical results indicate that LSTM consistently outperforms RNN in terms of Root Mean Squared Error (RMSE) and trend accuracy, especially during high volatility periods.

## I. INTRODUCTION

Stock price prediction is a challenging task that has captured the interest of both researchers and investors due to its potential financial rewards. Traditional statistical models, such as ARIMA, are often used for time series forecasting; however, they have limited capability in capturing complex temporal patterns in financial data. Recurrent Neural Networks (RNNs), particularly Long Short-Term Memory (LSTM) models, have demonstrated effectiveness in predicting time series data due to their ability to learn and remember long-term dependencies, which are crucial for forecasting stock price movements [1].



In this project, we aim to explore the effectiveness of RNN and LSTM models in predicting the closing prices of major technology companies, including Amazon, Apple, Microsoft, and Google. A key goal is to understand the capabilities and limitations of these models when applied to financial data, focusing on how their performance varies under different time

periods. This approach is necessary because stock prices are influenced by numerous factors that change over time, making it ineffective to use outdated data for current predictions.

## II. METHOD

### A. Problem Formulation

The primary objective is to predict the closing prices of Amazon, Apple, Microsoft, and Google stocks using RNN and LSTM models, and to compare their performance. The dataset consists of historical daily stock prices from January 2018 to January 2024, focusing specifically on the closing price as the target variable. The models utilize the past 2 and 4 years of closing prices to predict the next day's closing price. Model performance is evaluated using metrics such as Root Mean Squared Error (RMSE).

Two models were developed for comparison:

- **RNN Model:** A simple Recurrent Neural Network with one or two hidden layers.
- **LSTM Model:** A Long Short-Term Memory model with two LSTM layers (64 and 32 units) followed by dense layers.

### B. Methodologies

The models were implemented using Keras and trained with the following setup:

- **Input:** Sequences of 60 past closing prices.
- **Output:** Predicted closing price for the next day.
- **Training and Evaluation:** The dataset was split into training and testing sets (95% for training, 5% for testing). Models were trained for 20 epochs with a batch size of 32, and their performance was evaluated using RMSE and visualizations of predicted versus actual values.

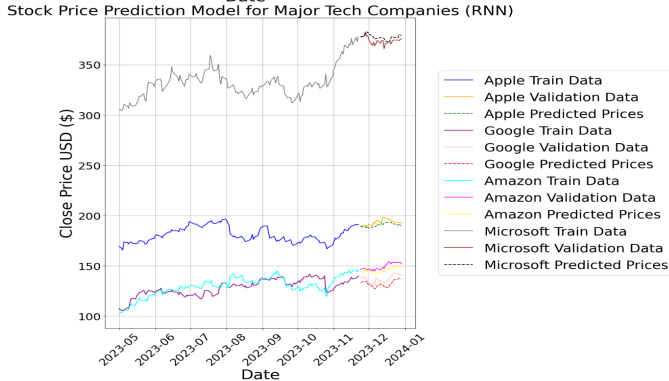
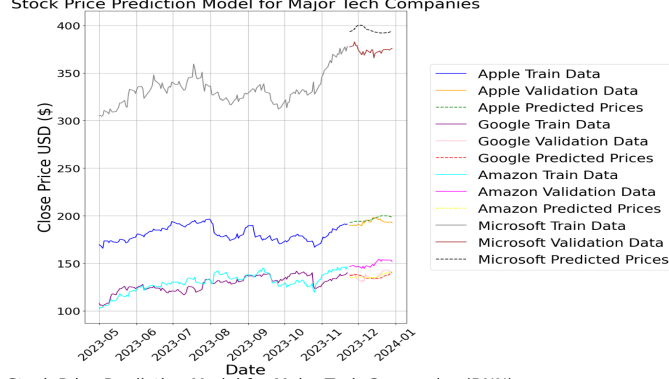
## III. RESULTS

### A. Data Pipeline and Model Setup

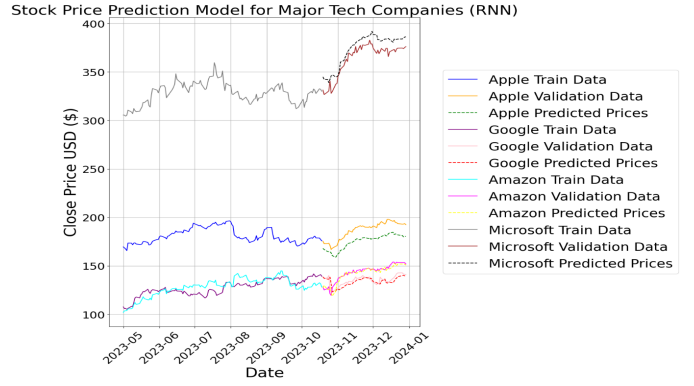
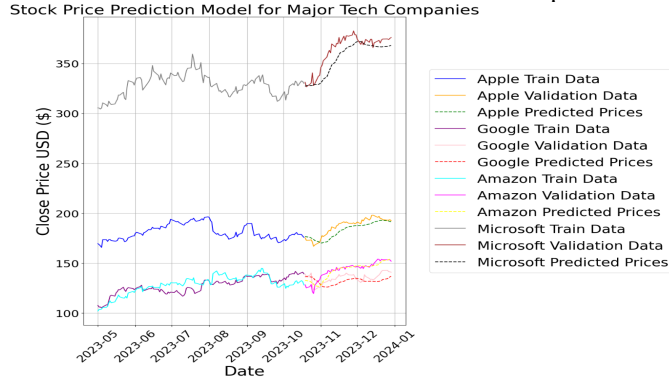
The data pipeline consisted of collecting stock price data, filtering it to include relevant time periods, and preparing the training and testing datasets. The LSTM model, trained with 64 and 32 units in two LSTM layers, was compared against the RNN model with respect to their ability to predict the closing price.

## B. Numerical Simulation Results

The LSTM model consistently outperformed the RNN model across all four companies. The following figure presents the comparison of actual and predicted closing prices for Apple, Google, Amazon, and Microsoft from January 2022 to January 2024. Specifically, the RMSE for LSTM are 2.561, 3.983, 10.665, and 3.858 while those for RNN are 6.475, 4.631, 13.926, and 17.594, respectively. It seems that for most cases, LSTM performs better. The figure indicates that the LSTM model was better at capturing the overall trend of the stock price, whereas the RNN model struggled, especially during high volatility periods for Microsoft and Apple. The RMSE for the LSTM model was significantly lower than that of the RNN model, indicating better predictive performance.



For the period of 4 years, the RMSE for LSTM are 4.356, 4.705, 4.649 and 8.535 while those for RNN are 6.124, 3.556, 4.298, and 16.256. From the figures, it seems that LSTM performs slightly better than the RNN method with more accurate predictions.



## C. Interpretation of Results

The numerical results showed that the LSTM model's RMSE was consistently lower than that of the RNN model across all companies, highlighting its superior ability to capture trends and minimize prediction errors. Specifically, during high volatility periods, the LSTM model's ability to retain critical information allowed it to provide more accurate predictions, whereas the RNN model struggled, leading to larger discrepancies.

The analysis also emphasized the importance of the time period chosen for training, as stock price trends evolve and models trained on outdated data may not effectively capture recent patterns. The LSTM model's ability to adapt to changing trends, as shown in the four-year RMSE comparison, suggests that it is a more robust choice for stock price prediction in dynamic financial markets.

## IV. CONCLUSION

This study compared RNN and LSTM models for stock price prediction of major technology companies. The LSTM model outperformed the RNN model in terms of both RMSE and visual tracking of the actual stock prices. The LSTM's ability to capture long-term dependencies made it more effective for this task. Additionally, the analysis emphasized the importance of selecting an appropriate training period for accurate prediction, given the highly dynamic nature of stock prices.

In future work, additional features such as trading volume or sentiment analysis could be incorporated to further improve model accuracy. Furthermore, more advanced models like Transformers could be explored to better capture long-term dependencies and complex relationships in financial data.

## REFERENCES

- [1] M. Waqas and U. W. Humphries, "A critical review of RNN and LSTM variants in hydrological time series predictions," *MethodsX*, vol. 13, p. 102946, 2024. doi: <https://doi.org/10.1016/j.mex.2024.102946>.